

Factors Influencing Vehicle Fuel Efficiency: An Analysis of Weight, Horsepower, and Displacement

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Introduction

Fuel efficiency is an important characteristic of vehicles because it directly affects fuel consumption, environmental impact, and the cost of transportation. As a vehicle becomes heavier or more powerful, it tends to use more fuel, though the precise nature of these relationships is worth examining more carefully through statistical analysis. Data for this analysis were obtained from the MPG dataset made publicly available through the UCI Machine Learning Repository, containing records on **392 vehicles** produced between 1970 and 1982.

Statistical modeling allows us to identify which vehicle features most strongly associates with fuel efficiency while accounting for other variables at the same time.

Primary Goal: This analysis examines the association between vehicle fuel efficiency and several mechanical characteristics using **multiple linear regression**, with the goal of identifying which predictors contribute meaningfully to explaining variation in **miles per gallon**.

The Variables include:

- Miles per gallon (mpg):** Fuel efficiency (response variable)
- Weight (lbs):** Total vehicle weight
- Horsepower:** Engine power output
- Displacement (cubic inches):** Total engine cylinder volume

Scientific Hypothesis:

It is hypothesized that **vehicle weight**, **horsepower**, and **engine displacement** are each negatively associated with fuel efficiency, and that at least one of these predictors remains significant after accounting for the others.

Methods

To examine the relationship between fuel efficiency and vehicle characteristics, **multiple linear regression models** were fitted to a dataset of **392 vehicles**, after removing observations with missing values for horsepower.

Model Specification

The initial regression model included three explanatory variables. Weight, horsepower, and displacement were selected as predictors because they are directly related to engine performance and vehicle size, and each is expected to influence how efficiently a vehicle uses fuel.

$$mpg = \beta_0 + \beta_1(weight) + \beta_2(horsepower) + \beta_3(displacement) + \epsilon$$

This model estimates expected fuel efficiency as a linear function of weight, horsepower, and engine displacement, holding the remaining predictors constant.

Model Comparison

To determine whether all three predictors were necessary, two **nested models** were compared. Because weight, horsepower, and displacement tend to move together in practice, including all three simultaneously risked introducing **multicollinearity**, where predictors explain overlapping variation in the response. Model 1:

$$mpg \sim weight + horsepower$$

Model 2:

$$mpg \sim weight + horsepower + displacement$$

An **ANOVA partial F-test** was used to formally evaluate whether adding displacement significantly improved model fit beyond what weight and horsepower already provided.

Model Diagnostics

- Several diagnostic tools were used to check whether the assumptions of linear regression were reasonably satisfied:
- Residuals vs fitted plots** to assess linearity
 - Q-Q plots** to evaluate normality of residuals
 - Scale-location plots** to examine constant variance
 - Leverage values and Cook's distance** to detect influential observations

These diagnostics were used to determine whether the regression model provides a reliable representation of the data.

Results

Exploratory Analysis

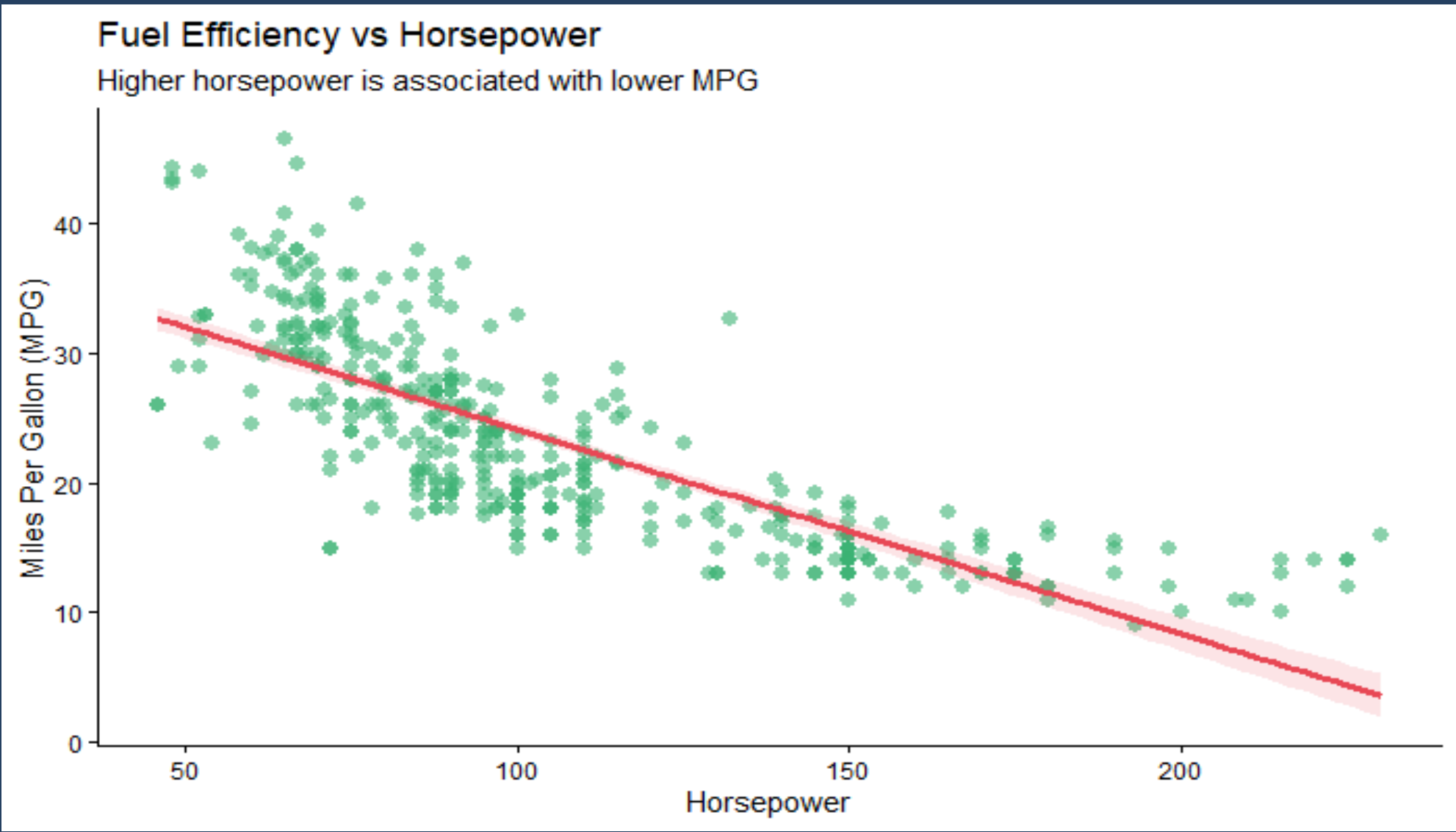
Scatterplots were used to look at the relationship between fuel efficiency and each potential predictor before any modeling was done. **Summary statistics** for all variables are reported in (See table 1).

The exploratory analysis suggested:

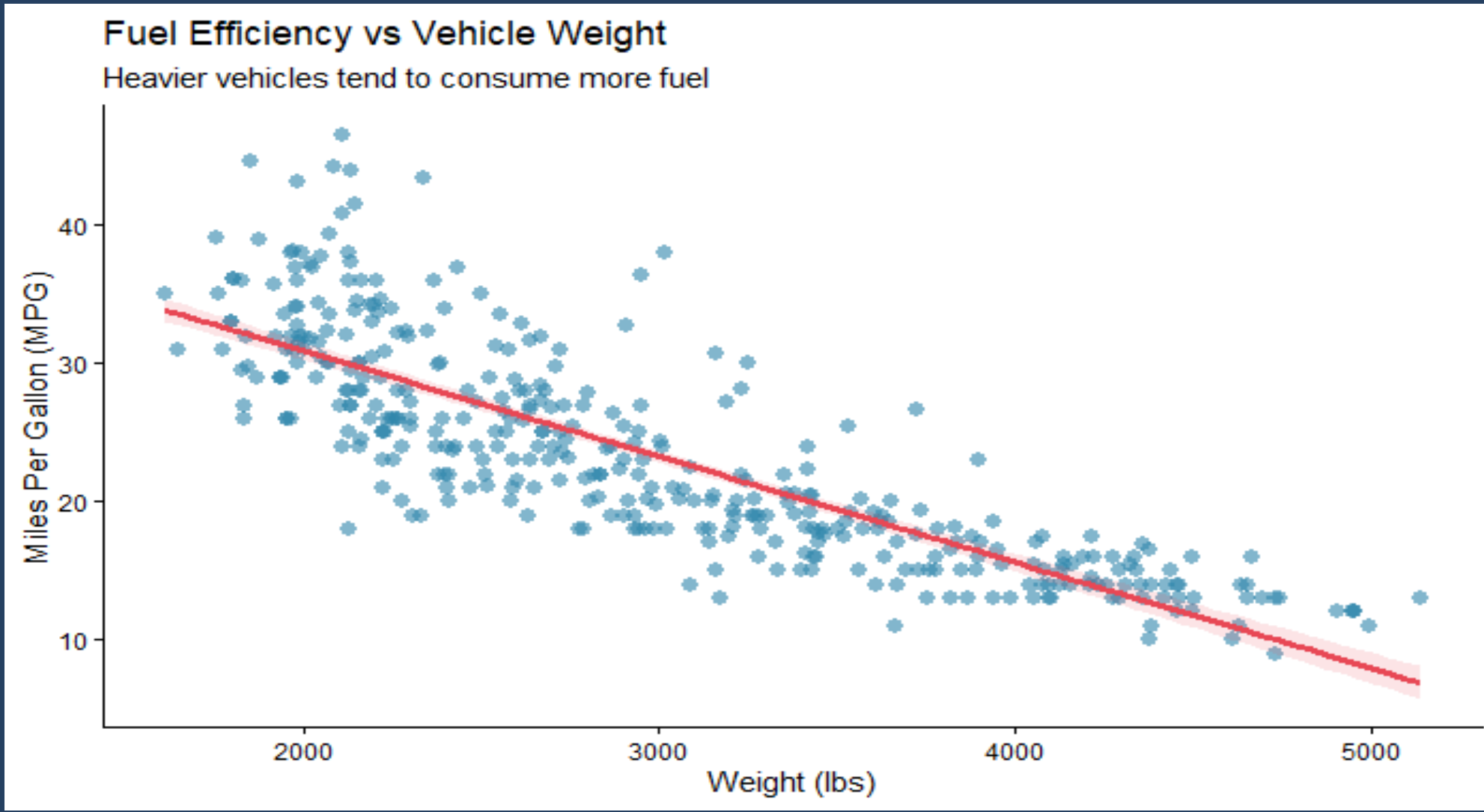
- Vehicle weight** is negatively associated with mpg, with heavier vehicles tending to show noticeably lower fuel efficiency (See figure 1)
- Horsepower** also shows a clear negative relationship with mpg, where more powerful engines tend to consume more fuel (See figure 2)
- Engine displacement** appears negatively related to mpg as well, though the relationship looks weaker and less distinct compared to the other two predictors (See figure 3)

These patterns supported the use of a **multiple linear regression** model to quantify these associations while controlling for each predictor simultaneously.

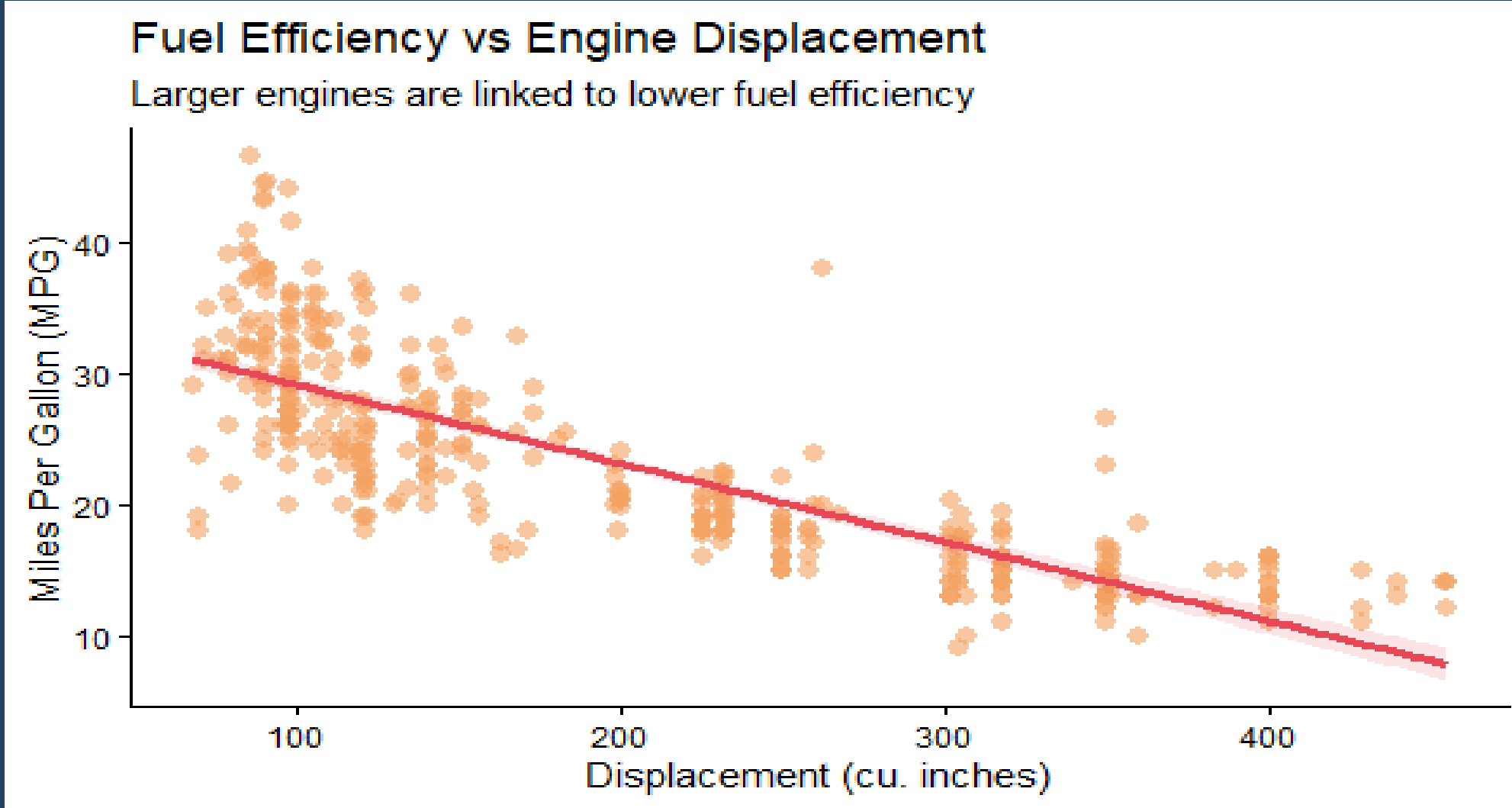
(Figure 1)



(Figure 2)



(Figure 3)



(Table 1)

Variable	Mean	Median	Min	Max
Miles per Gallon (MPG)	23.45	22.75	9.00	46.60
Vehicle Weight	2978	2804	1613	5140
Horsepower	104.5	93.5	46.0	230.0
Engine Displacement	194.4	151.0	68.0	455.0

Comparison Results

The **ANOVA partial F-test** comparing the two nested models produced an **F-statistic** of 0.768 and a **p-value** of 0.381 (See table 2). This indicates that adding engine displacement does not significantly improve model fit once weight and horsepower are already included. The result is consistent with the idea that displacement shares a great deal of overlap with the other two predictors, offering little additional explanatory power on its own. Based on this result, the **reduced model** containing only weight and horsepower was selected as the final model:

$$mpg = \beta_0 + \beta_1(weight) + \beta_2(horsepower) + \epsilon$$

(Table 2)

Model	Residual DF	RSS	F-statistic	p-value
mpg ~ weight + horsepower	389	6993.8		
mpg ~ weight + horsepower + displacement	388	6980.0	0.768	0.381

Regression Results

The estimated regression equation is (See table 3):

$$mpg = 45.64 - 0.00579(weight) - 0.0473(horsepower)$$

Interpretation of coefficients:

- Weight coefficient (-0.00579)**
Holding horsepower constant, an increase of 1 pound in vehicle weight is associated with a decrease of approximately 0.0058 mpg.
- Horsepower coefficient (-0.0473)**
Holding weight constant, an increase of 1 unit of horsepower is associated with a decrease of approximately 0.047 mpg.

Both predictors were **statistically significant** with **p-values** less than 0.001, providing strong evidence that weight and horsepower are each meaningfully associated with fuel efficiency after accounting for one another.

The model explains a large portion of variability in fuel efficiency

- R² = 0.706**
- Adjusted R² = 0.705**

This means that approximately **70% of the variation in mpg** is explained by weight and horsepower.

(Table 3)

Predictor	Estimate	Std Error	t value	p-value
Intercept	45.64	0.793	57.54	<0.001
Weight	-0.00579	0.00050	-11.54	<0.001
Horsepower	-0.0473	0.0111	-4.27	<0.001

Results Cont.

Model Diagnostics

Diagnostic plots were examined to check whether the assumptions of linear regression were adequately met by the final model.

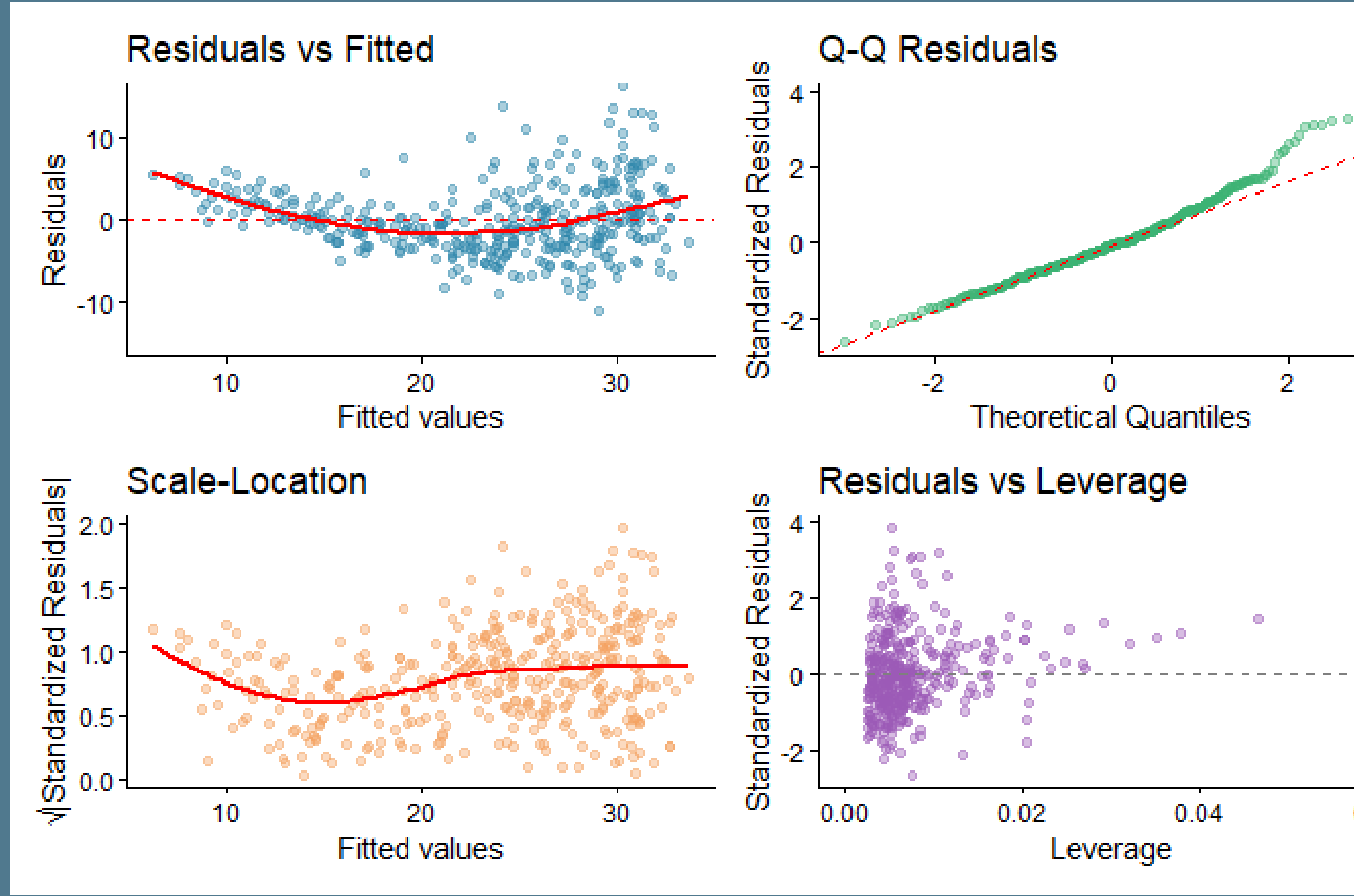
Residuals vs Fitted Plot: Shows no strong systematic pattern, suggesting the **linearity assumption** is reasonable.

Q-Q Plot: Indicates the residuals were approximately **normally distributed**, with only minor deviations present in the extreme tails.

Scale-location Plot: Suggests a mild increase in residual variance for larger fitted values, indicating slight **heteroscedasticity**. However, the effect appears small and does not substantially affect interpretation of the model.

Residual vs Leverage Plot: **Leverage values** and **Cook's distance** were examined to identify potentially influential observations. The largest Cook's distance was approximately 0.037, well below the commonly used threshold of 0.5, indicating that no single observation has a disproportionate influence on the fitted model.

Overall, the diagnostic analysis suggests that the regression assumptions are adequately satisfied and that the final model provides a reliable fit to the data.



Conclusion

This analysis looked at the relationship between vehicle characteristics and fuel efficiency using **multiple linear regression** applied to a dataset of **392 vehicles** produced between 1970 and 1982.

The results indicate that:

- Vehicle weight** is strongly associated with lower fuel efficiency
- Higher horsepower** is also associated with lower miles per gallon
- Engine displacement** does not significantly improve the model once weight and horsepower are included, and was excluded from the final model based on the **partial F-test** result (p = 0.381)

The final regression model explains approximately 70% of the variation in **fuel efficiency**, indicating that these mechanical characteristics play an important role in determining fuel consumption. A few limitations are worth noting. Slight **heteroscedasticity** was detected in the residual plots, which may affect the reliability of predictions at higher fitted values. Additionally, because the data are **observational**, the results reflect associations between variables rather than causal relationships. It cannot be concluded from this analysis that changes in weight or horsepower directly cause changes in fuel efficiency. The findings suggest that **heavier vehicles** and vehicles with **greater engine power** tend to have substantially lower fuel efficiency, and that engine size alone does not add meaningful predictive information beyond what weight and horsepower already capture.

Acknowledgements

Data obtained from the MPG dataset originally compiled by Carnegie Mellon University and made publicly available through the UCI Machine Learning Repository. Analysis conducted using R version 4.4.1

Data sources: <https://raw.githubusercontent.com/mwaskom/seaborn-data/master/mpg.csv>